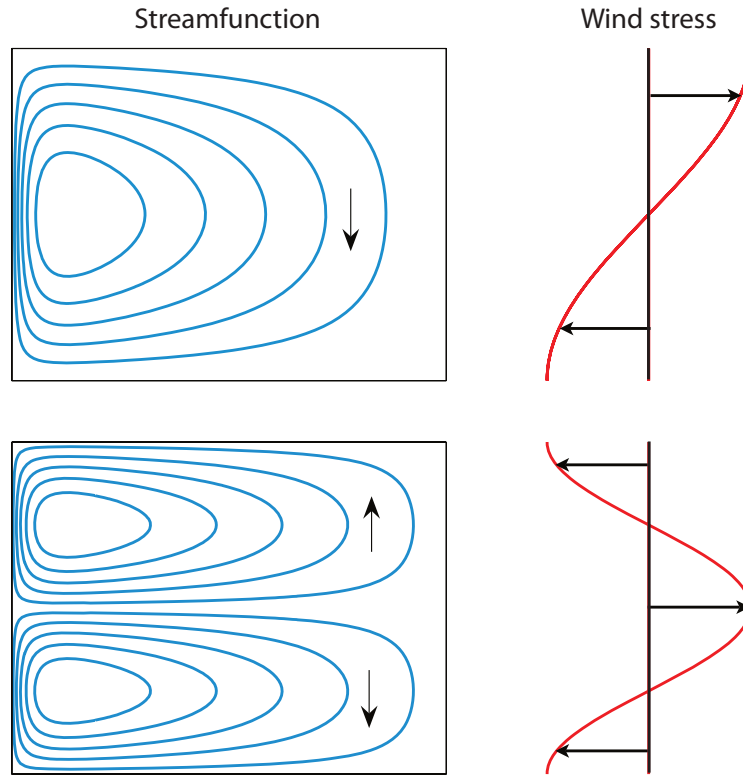


Fig. 14.4: Two solutions of the Stommel model. Upper panel shows the streamfunction of a single-gyre solution, with a wind stress proportional to $-\cos(\pi y/a)$ (in a domain of side a), and the lower panel shows a two-gyre solution, with wind stress proportional to $\cos(2\pi y/a)$. In both cases $\epsilon_S = 0.04$.



It is a relatively straightforward matter to generalize to other wind stresses, provided these also vanish at the two latitudes between which the solution is desired. It is left as a problem to show that in general

$$\psi_I = \int_{x_E}^x \text{curl}_z \boldsymbol{\tau}(x', y) dx', \quad (14.35)$$

and that the composite solution is

$$\psi = \psi_I - \psi_I(0, y) e^{-x/(x_E \epsilon_S)}. \quad (14.36)$$

14.3.3 The Munk Problem: Using Viscosity Instead of Drag

A natural variation on the Stommel problem is to use a harmonic viscosity, $\nu \nabla^2 \zeta$, in place of the drag term $-r\zeta$ in the vorticity equation, the argument being that the wind-driven circulation does not reach all the way to the ocean bottom so that an Ekman drag is not appropriate. This variation is called the 'Munk problem' or 'Munk model'. The problem is to find and understand the solution to the (dimensional) equation

$$\beta \frac{\partial \psi}{\partial x} = \text{curl}_z \boldsymbol{\tau}_T + \nu \nabla^2 \zeta = \text{curl}_z \boldsymbol{\tau}_T + \nu \nabla^4 \psi \quad (14.37)$$

in a given domain, for example a square of side a . Because the equation is of higher order we need two boundary conditions at each wall to solve the

Stommel and Munk Models of the Wind-Driven Circulation

Formulation

- Vertically integrated planetary-geostrophic equations, or a homogeneous fluid with nonlinearity neglected.
- Friction parameterized by a linear drag (Stommel model) or a harmonic Newtonian viscosity (Munk model) or both (Stommel–Munk model).
- Flat bottomed ocean.

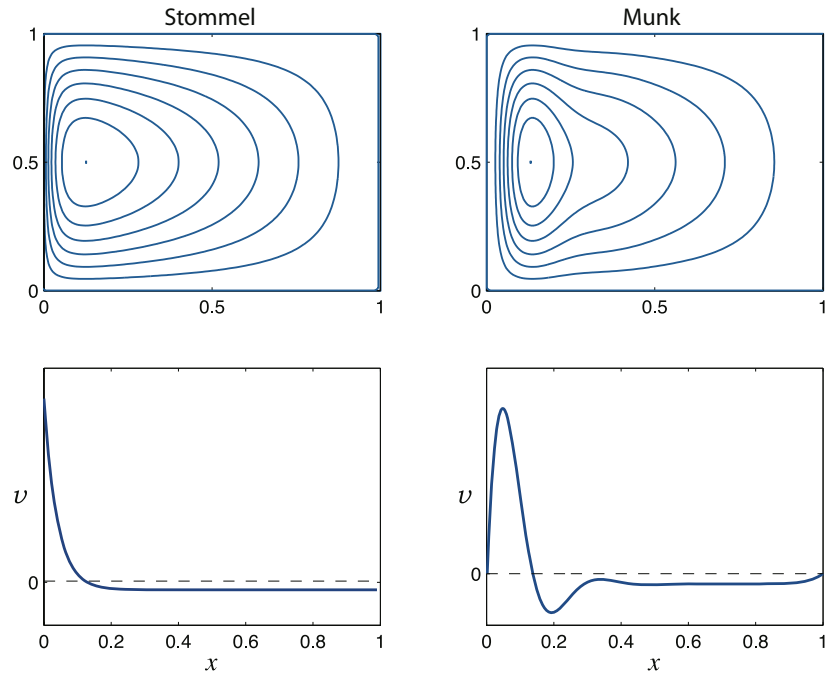
Properties

- The transport in the Sverdrup interior is equatorward for an anti-cyclonic wind-stress curl. This transport is exactly balanced by the poleward transport in the western boundary layer.
- There must be a boundary layer to satisfy mass conservation, and this must be a *western* boundary layer if the friction acts to provide a force of opposite sign to the motion itself. As there is a balance between friction and the β -effect, it is a ‘frictional boundary layer’. The western location does not depend on the sign of the wind stress, nor on the sign of the Coriolis parameter, but it does depend on the sign of β , and so on the direction of rotation of the Earth.
- In the Stommel model the boundary layer width arises by noting that the terms $r\nabla^2\psi$ and $\beta\partial\psi/\partial x$ are in approximate balance in the western boundary layer, implying boundary-layer scale of $L_S = (r/\beta)$. If r , the inverse frictional time, is chosen to be $1/20 \text{ days}^{-1}$ then $L_S \approx 60 \text{ km}$, similar to the width of the Gulf Stream. Unless the wind has a special form the Sverdrup flow is non-zero on the zonal walls and there must also be boundary layers there, but they are weaker and less visible.
- In the Munk model the balance in western boundary layer is between $\nu\partial^4\psi/\partial x^4$ and $\beta\partial\psi/\partial x$, implying a scale of $L_M = (\nu/\beta)^{1/3}$. There are also weak boundary layers on the eastern and zonal walls, to satisfy the no-slip (or free slip) condition.

problem uniquely, and as before for one of them we choose $\psi = 0$ to satisfy the no-normal-flow condition. For the other condition it is common to use a no-slip condition; that is $\psi_n = 0$ where the subscript denotes the normal derivative of the streamfunction, so that, for example, at $x = 0$ and $x = a$ we have $v = 0$.

Just as for the Stommel problem, the solution to (14.37) may be found by boundary-layer methods and the solution so found is very similar to that of the Stommel problem — the interior is almost identical, being given by the Sverdrup interior flow, and a boundary layer occurs in the west. The boundary layer now brings the alongshore flow to zero, as well as the normal flow, and it has a different thickness to that of the Stommel

Fig. 14.5: The Stommel and Munk solutions, (??) with $\epsilon_S = \epsilon_M^{1/3} = 0.04$, with the wind stress $\tau = -\cos \pi y$, for $x, y \in (0, 1)$. Upper panels are contours of streamfunction in the x - y plane, and the flow is clockwise. The lower panels are plots of meridional velocity, v , as a function of x , in the centre of the domain ($y = 0.5$). The Munk solution can satisfy both no-normal flow and one other boundary condition at each wall, here chosen to be no-slip.



problem. This thickness can be estimated by a scale analysis of (14.37), since in the boundary layer we expect the dominant balance to be between the beta term and the frictional term, whence, in a western boundary layer

$$\beta \frac{\partial \psi}{\partial x} \sim \nu \frac{\partial^4 \psi}{\partial x^4}, \quad (14.38)$$

and the thickness of the boundary layer, L_M is given by

$$L_M = \left(\frac{\nu}{\beta} \right)^{1/3}. \quad (14.39)$$

However, even though the frictional term in the Munk problem resembles a molecular viscosity the solution should not be regarded as being more realistic than the Stommel model. Rather, the similarity of the two model solutions attests the *qualitative* realism of both models and the relative insensitivity of the solution to the precise form of the friction, but neither model is quantitatively accurate.

14.3.4 Application to the North Atlantic

It is all very well to produce an intense western boundary current in a square domain, but another matter to produce a flow field that actually resembles the real ocean. To see if this can be done we solve the Stommel–Munk problem in a domain with realistic geometry, albeit a flat-bottomed domain with no topography, using the observed winds to drive the flow.

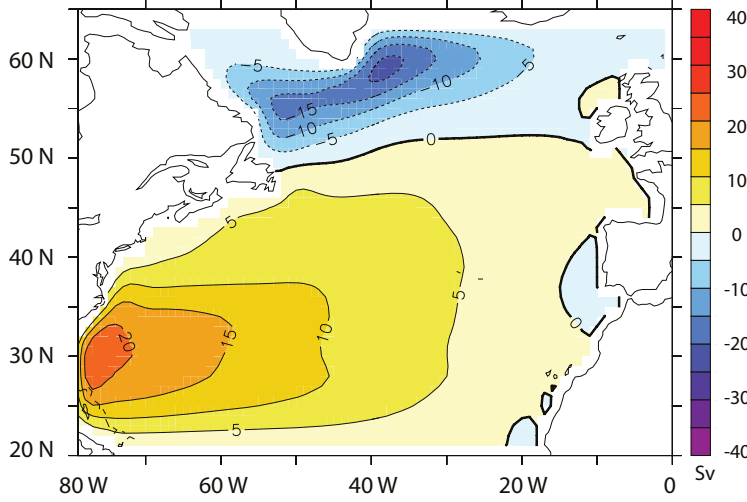


Fig. 14.6: The solution (streamfunction, in Sverdrups) to the Stommel–Munk problem numerically calculated for the North Atlantic, using the observed wind field. The calculation reproduces the observed large-scale patterns, but compared to observations the separation of the Gulf Stream from the coast is a little too far north and the Gulf Stream extension is too diffuse. Compare with Figs. 14.1 and 14.3.1.

The solution must be obtained numerically, but the equation we solve is equivalent to

$$\beta \frac{\partial \psi}{\partial x} = \text{curl}_z \tau_T - r\zeta + \nu \nabla^2 \zeta = \text{curl}_z \tau_T + \nu \nabla^4 \psi, \quad (14.40)$$

and the solution is illustrated in Fig. 14.6.

The solution captures many of the features of the observed flow — notably the subtropical and subpolar gyres and the Gulf Stream itself — although it is quantitatively deficient in some aspects. For example, only the current in the west is intensified, and not the Gulf Stream extension as it flows eastward, and there is insufficient recirculation in the western part of the gyre. These failings can be ascribed in various ways to the lack of nonlinearity, stratification and bottom topography in the model problem, but overall the model provides a compelling explanation of some of the main features of the ocean circulation.

14.4 EFFECTS OF NONLINEARITY

Nonlinearity affects the ocean circulation in a variety of ways. One is that the ocean, like the atmosphere, is baroclinically unstable and the resulting baroclinic eddies lead to weather in the ocean in a similar way to that giving weather in the atmosphere. But even if we just consider the depth integrated or barotropic flow then nonlinearity can be important, as we now explore.

14.4.1 A Nonlinear Wind-Driven Model

Let us suppose that the ocean satisfies the barotropic vorticity equation, namely

$$J(\psi, \zeta) + \beta \frac{\partial \psi}{\partial x} = \text{curl}_z \tau_T - r\zeta. \quad (14.41)$$